Winning Array Designs, Per Myopia Level

Visualized to Scale, with Fabrication Recommendation

IYPT Pinhole Sunglasses — Array Geometry Sub-Study

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1 Summary Table

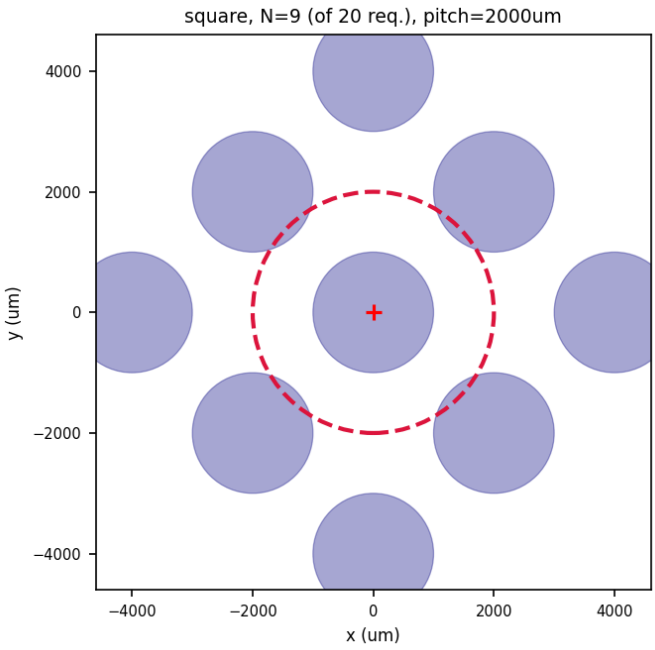
All designs below are deterministic (no seed dependence) and confirmed seed-robustness-clean, per the final corrected evaluation run. Blue circles are pinholes, drawn to true relative scale; the dashed red circle is the 4 mm pupil at straight-ahead gaze ($\varphi = 0$).

M (D)	Family	Spacing	Ghost/Vign. margin	Blind rate	Eff. density
0	ring_offset_concentric	rs=1000 μm, N=11	1.000 / 1.000	15.6%	3.60e-8
1	**square**	**pitch=2000, N=20**	**0.128 / 0.128**	**8.2%**	**4.16e-8**
2–3	ring_offset_concentric	rs=4000 μm, N=20	$+\infty$ / $+\infty$	$\sim$10.7%	5.89e-9
4–5	ring_offset_concentric	rs=4000 μm, N=20	$+\infty$ / $+\infty$	$\sim$10.5%	6.02e-9
6, 8	ring_offset_concentric	rs=4000 μm, N=20	$+\infty$ / $+\infty$	10.1–10.3%	6.1–6.2e-9
global	hexagonal	pitch=2000, 30°, N=7	$+\infty$ / $+\infty$	17.3%	1.81e-8

Note

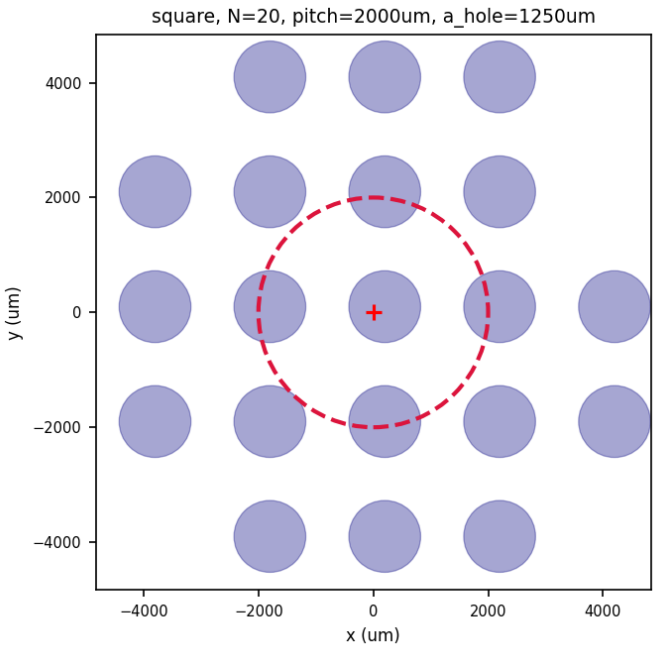
$+\infty$ margin means zero simultaneously-illuminated hole pairs at every tested gaze angle — a structural, geometric guarantee rather than a narrowly-cleared numeric threshold. M=1's finite margin (0.128) is the one design in this set operating with a real, measured safety buffer rather than a structural impossibility of ghosting.

2 M = 0 D



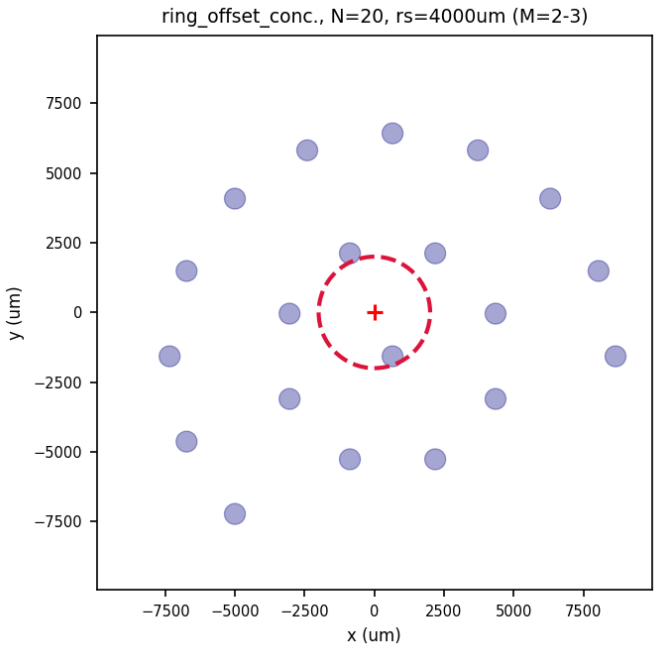
`ring_offset_concentric`, ring_spacing=1000 μm, N=11, a_hole=2000 μm. Margin 1.000/1.000 — passes with real (not structural) margin. Blind rate 15.6%.

3 M = 1 D



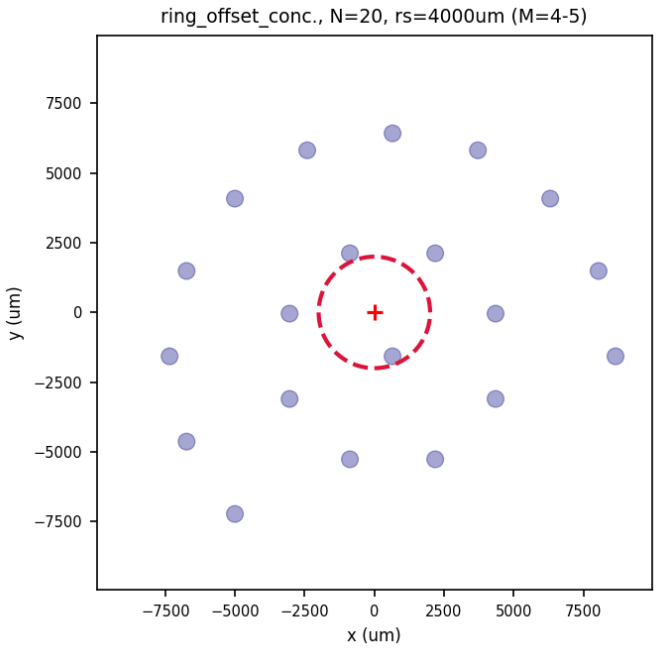
`square`, pitch=2000 μm, N=20, a_hole=1250 μm. Margin 0.128/0.128 — the one finite-margin design in this set. Blind rate 8.2%, the lowest of any winner.

4 M = 2–3 D



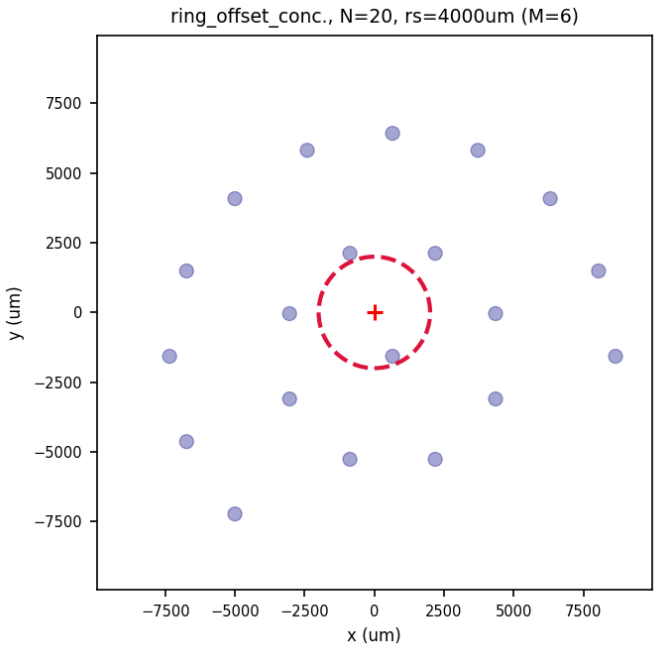
`ring_offset_concentric`, ring_spacing=4000 μm, N=20, a_hole=750 μm. Margin $+\infty$ (structurally ghost-free). Blind rate $\sim$10.7%.

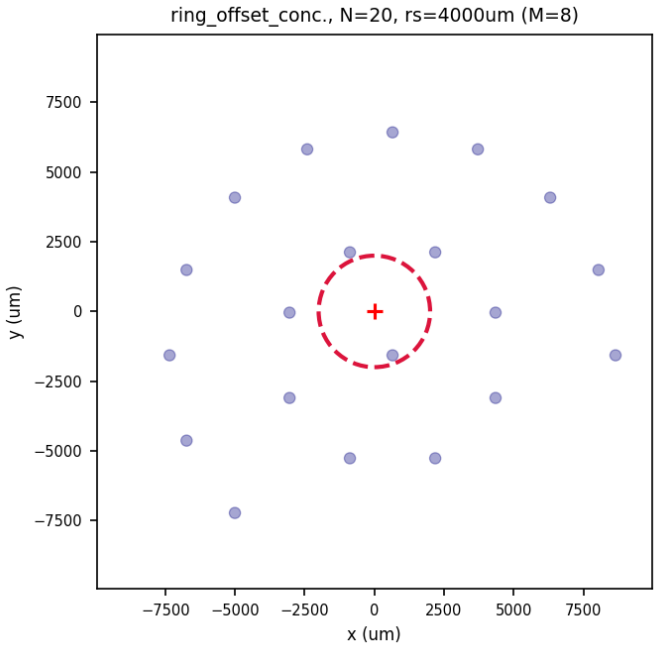
5 M = 4–5 D



`ring_offset_concentric`, ring_spacing=4000 μm, N=20, a_hole=600 μm. Same geometry as M=2–3, smaller hole diameter (per the acuity-optimal chart). Margin $+\infty$. Blind rate $\sim$10.5%.

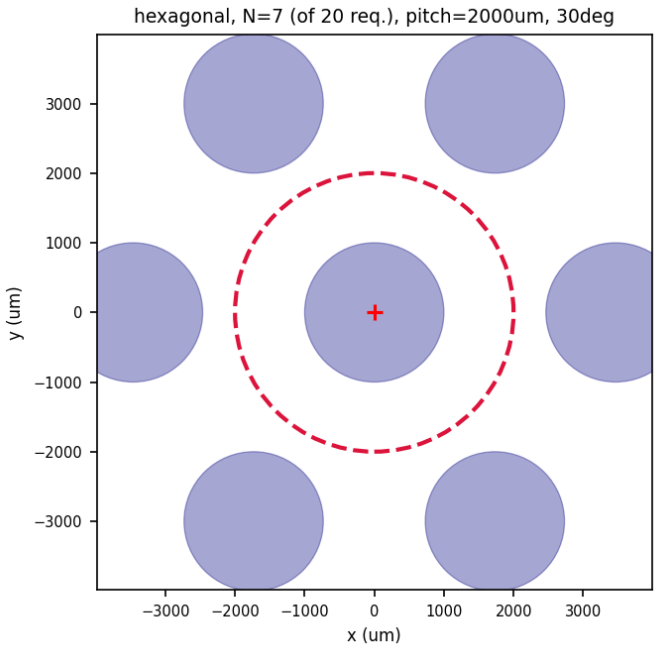
6 M = 6 D, M = 8 D





Same `ring_offset_concentric` geometry again, a_hole=500 μm (M=6, left) and 400 μm (M=8, right). Margin $+\infty$ at both. Blind rate 10.3% and 10.1% respectively.

7 Global Scope (worst-case across all M)



`hexagonal`, pitch=2000 μm, rotation=30°, N=7, a_hole=2000 μm. Margin $+\infty$. Blind rate 17.3%, the highest of any winner — the cost of guaranteeing safety across the full M=0–8D range with one fixed geometry.

8 Which to Use for Fabrication

Nine full designs is more than a first physical test needs. Two candidates cover the practically useful range and are the ones this project recommends building first.

Recommended for Fabrication

1. `square`, pitch=2000 μm, N=20, a_hole=1250 μm (M=1). This is the single most informative design to test physically: it is the only winner with a real, finite, measured safety margin rather than a structural zero-active-pair guarantee. A physical measurement here directly tests whether the model's finite-margin prediction (0.128) holds up in reality — the structural-safety designs cannot tell you this, since they are safe by geometry regardless of small modeling errors.

2. `ring_offset_concentric`, ring_spacing=4000 μm, N=20 (M=2 through 8). One physical frame, six aperture-size inserts (750, 600, 500, 400 μm) covers the entire higher-myopia range with a single hole pattern. Lowest fabrication cost per M-level covered of any option in the table.

8.1 Why Not the Others First

- **M=0's ring_offset_concentric** and **global's hexagonal** are both structurally safe with wide margin — lower priority for a first physical test than the finite-margin M=1 design, since they are less likely to reveal a gap between prediction and reality.
- All designs in this table are already the survivors of an extensive audit (density-metric bias, boundary blackout cases, and a seed-lottery defect were each found and fixed against earlier candidates) — none of the remaining seven "not recommended first" designs carry a known defect, they are simply lower priority for the first fabrication round.

8.2 Practical Note

The M=1 design (1250 μm holes) and the M=2–8 design family (750–400 μm holes) together span the full aperture range needed for laser-cut shim fabrication testing described elsewhere in this project — test the smallest hole size (400 μm, M=8 insert) first to confirm cutting-edge quality before committing material to the full set.